Communication Systems Lecture-9: Line Codes and Spectra

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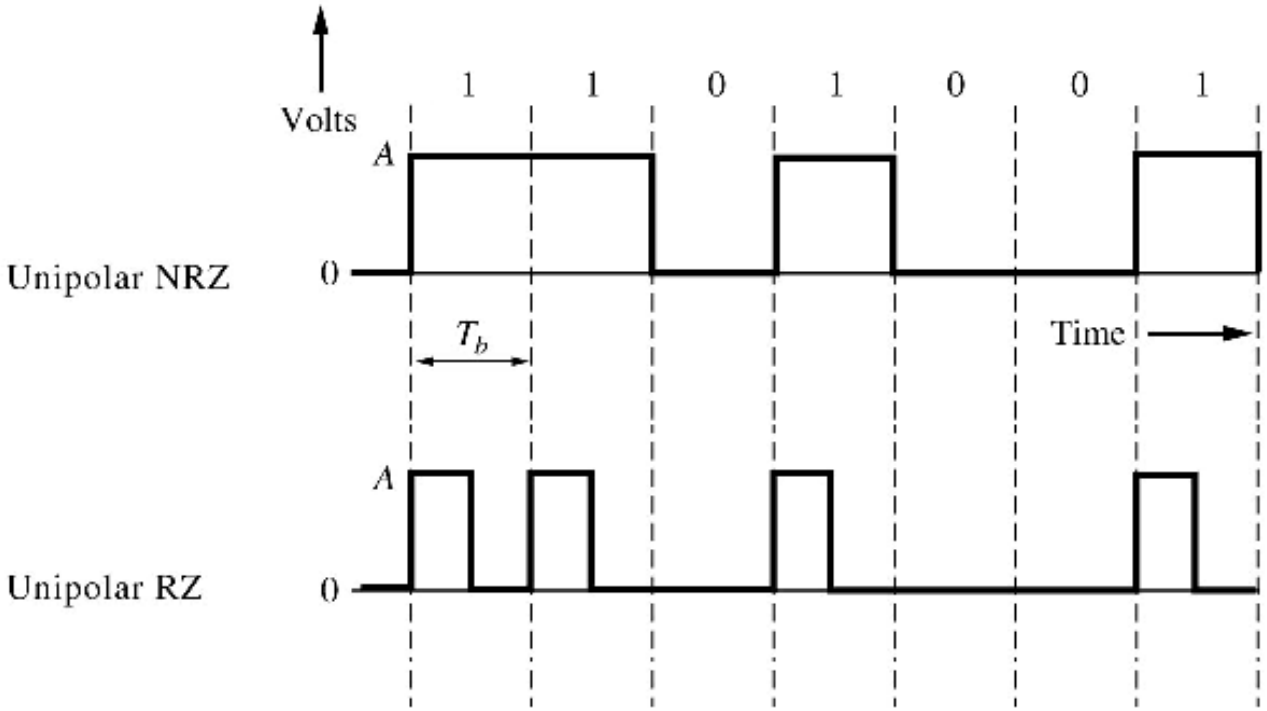
Binary Line Codes (1)

- **Binary Line Code**: corresponds to the representation of binary 0's and 1's with various serial-bit signaling formats.
- There are two major categories of line codes:
  - return-to-zero (RZ):
    - The waveform returns to zero-volt level for a portion of the bit interval (usually one half of the bit interval).
  - nonreturn-to-zero.

Binary Line Codes (2)

Unipolar Signaling:

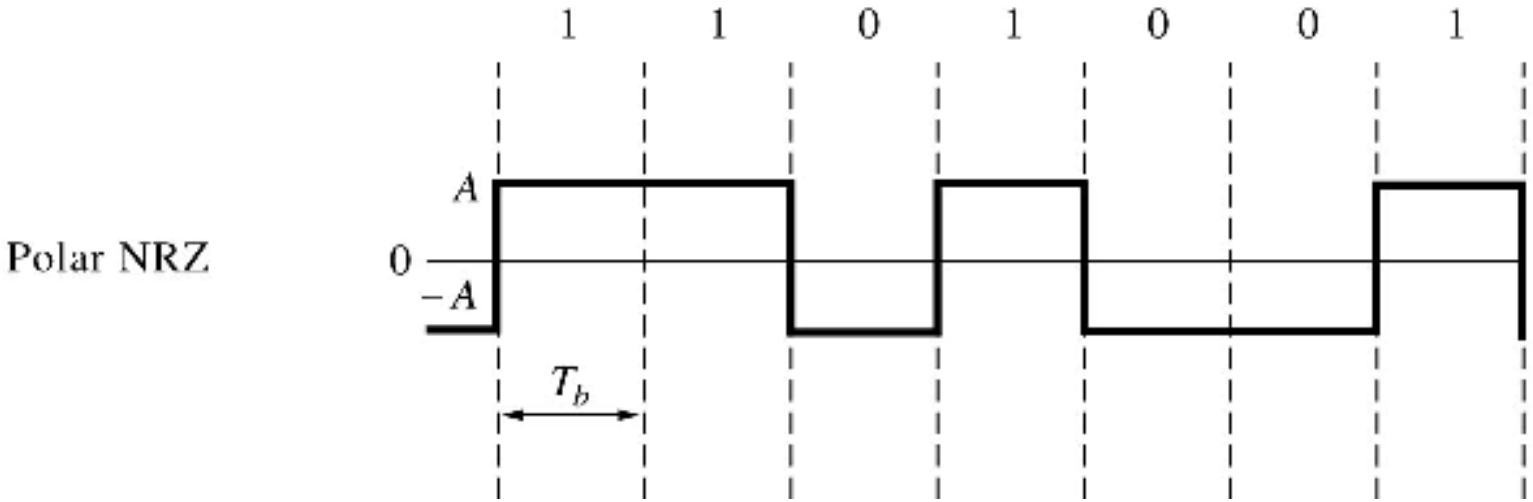
- A binary 1 is represented by a high level ($+A$).
- A binary 0 is represented by a zero level.



Binary Line Codes (3)

Polar Signaling:

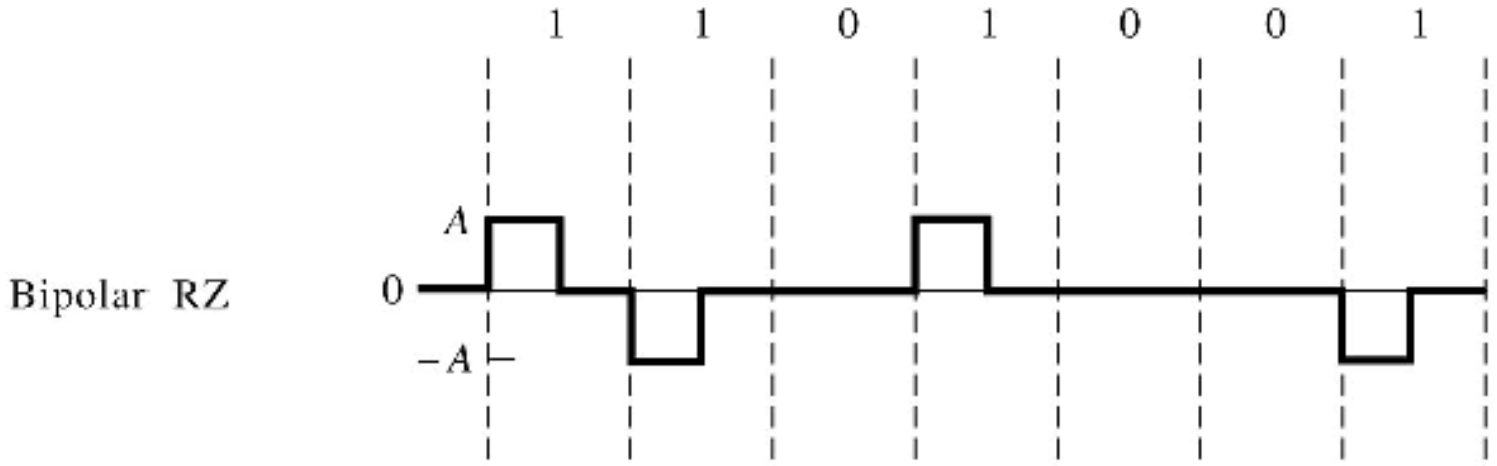
- A binary 1 is represented by a positive level ($+A$).
- A binary 0 is represented by a negative level having the same absolute value ($-A$).



Binary Line Codes (4)

Bipolar (Pseudoternary) Signaling:

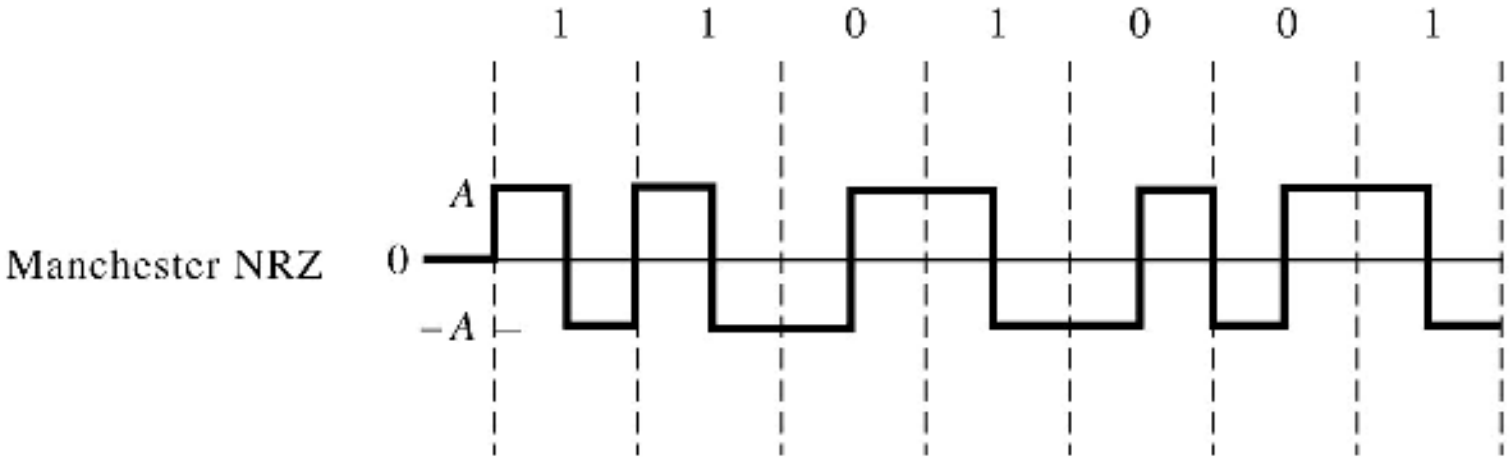
- Binary 1's are represented by alternately positive or negative values.
- A binary 0 is represented by a zero level.
- The term pseudoternary refers to the use of three levels to represent two-level (binary) data.
- This is also called Alternate Mark Inversion (AMI).



Binary Line Codes (5)

Manchester Signaling:

- A binary 1 is represented by a positive half-bit period pulse followed by a negative half-bit period pulse.
- A binary 0 is represented by a negative half-bit period pulse followed by a positive half-bit period pulse.



Desirable properties of line codes (1)

Self-Synchronization:
There must be enough timing information built into the code so that the clock signal can be easily recovered. A long series of 0's or 1's should not cause a problem in time recovery.

- Unipolar NRZ: Timing can not be recovered for long series of 0's or 1's.
- Unipolar RZ: Timing can not be recovered for long series of 0's.
- Polar NRZ: Timing can not be recovered for long series of 0's or 1's.
- Bipolar RZ: Timing can not be recovered for long series of 0's.
- Manchester code: Timing can always be recovered.

Desirable properties of line codes (2)

A spectrum that is suitable for the channel:
A channel is said to be DC coupled if its frequency response extends to $f = 0$. Otherwise, it is said to be AC coupled.

- Unipolar codes: They have a non-zero DC value. They are not suitable for AC coupled channels.
- Polar codes: When the probability of transmitting a binary 0 is the same as that of transmitting a binary 1, the DC component is equal to zero.
- Bipolar code: The DC component always has a negligible value.
- Manchester code: The DC component is always equal to zero.

Desirable properties of line codes (3)

Bandwidth:

- The signal bandwidth must be as small as possible.
- The signal bandwidth must be small compared to the channel bandwidth. Otherwise, the system will suffer from excessive inter-symbol-interference (as will be discussed later).
- In general, the bandwidth of RZ codes is double the bandwidth of NRZ codes (since the rectangular pulses of RZ codes are limited to one half of the period duration).

Cost:

- Unipolar codes require only one power supply (resulting in a low cost).
- Polar, bipolar and Manchester codes require two power supplies (one for each non-zero amplitude level).

Power Spectra for Binary Line Codes (1)

- A digital signal (or line code) can be represented by:

$$s(t) = \sum_{n=-\infty}^{+\infty} a_n f\left(t - nT_s\right)$$

where:

  - $f(t)$ stands for the pulse shape.
  - $\{a_n\}$ is the set of random data.
  - T_s is the duration of one symbol:
    - $T_s = T_b$ for binary signaling.
    - $T_s = lT_b$ for multilevel signal with $L = 2^l$ levels.
- Example: for unipolar NRZ:
  - $f(t) = \mathrm{rect}\left(\frac{t}{T_b}\right)$.
  - $a_n = +A$ for a binary 1.
  - $a_n = 0$ for a binary 0.

Power Spectra for Binary Line Codes (2)

- It can be shown that the PSD of a digital signal is given by:

$$\mathcal{P}_s(f) = \frac{|F(f)|^2}{T_s} \sum_{k=-\infty}^{+\infty} R(k) e^{j2\pi k f T_s}$$

where:

- $F(f)$ is the Fourier transform of $f(t)$ and T_s is the duration of one symbol.
- $R(k)$ is the autocorrelation of the data:

$$R(k) = \sum_{i=1}^{I} (a_n a_{n+k})_i \, P_i$$

where:

- a_n and a_{n+k} are the levels of the n-th and $(n+k)$-th symbols respectively.
- P_i is the probability of having the i-th $a_n a_{n+k}$ product.

- The PSD of digital signals depends on:
  1. The pulse shape used.
  2. The statistical properties of the data.

PSD for Unipolar NRZ Signaling (1)

- For unipolar signaling, the possible values of the a's are $+A$ and 0. In what follows we assume that:
  - These values are equally likely.
  - The data are independent.
- First, we evaluate $R(k)$. For $k = 0$, the possible values of $a_n a_{n+k} = a_n^2$ are:

$$A \times A = A^2 \quad \text{with probability} \quad \frac{1}{2}$$

$$0 \times 0 = 0 \quad \text{with probability} \quad \frac{1}{2}$$

- Consequently:

$$R(0) = \sum_{i=1}^{I=2} (a_n a_n)_i \, P_i$$

$$= A^2 \frac{1}{2} + 0 \frac{1}{2} = \frac{1}{2} A^2$$

PSD for Unipolar NRZ Signaling (2)

- For $k \neq 0$, the possible values of $a_n a_{n+k}$ are $A \times A$, $A \times 0$, $0 \times A$ and 0×0. They all occur with a probability of $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. Consequently:

$$R(k) = \sum_{i=1}^{I=4} (a_n a_{n+k})_i \, P_i$$
$$= A^2 \frac{1}{4} + 0\frac{1}{4} + 0\frac{1}{4} + 0\frac{1}{4} = \frac{1}{4} A^2$$

- On the other hand, for NRZ the pulse shape and its Fourier transform are given by:

$$f(t) = \mathrm{rect}\left(\frac{t}{T_b}\right) \quad ; \quad F(f) = T_b \mathrm{sinc}\,(T_b f)$$

PSD for Unipolar NRZ Signaling (3)

- Since we are considering binary signaling: $T_b = T_s$. The PSD of the unipolar NRZ line code is:

$$\mathcal{P}_s(f) = \frac{1}{T_b} T_b^2 \mathrm{sinc}^2\left(T_b f\right) \left[\frac{A^2}{2} + \frac{A^2}{4} \sum_{k=-\infty\ ,\ k\neq 0}^{+\infty} e^{j2\pi k f T_b}\right]$$

$$= T_b \mathrm{sinc}^2\left(T_b f\right) \left[\frac{A^2}{4} + \frac{A^2}{4} \sum_{k=-\infty}^{+\infty} e^{j2\pi k f T_b}\right]$$

$$= \frac{A^2 T_b}{4} \mathrm{sinc}^2\left(T_b f\right) \left[1 + \sum_{k=-\infty}^{+\infty} e^{j2\pi k f T_b}\right]$$

PSD for Unipolar NRZ Signaling (4)

- On the other hand, we previously proved that:

$$\mathcal{F}\left[\sum_{n\in\mathbb{Z}}\delta(t-nT_b)\right]=\frac{1}{T_b}\sum_{n\in\mathbb{Z}}\delta\left(f-\frac{n}{T_b}\right)$$

- In the same way, $\mathcal{F}[\delta(t)]=1$ implying that $\mathcal{F}[\delta(t-nT_b)]=e^{-j2\pi nT_bf}$. Therefore:

$$\begin{aligned}\mathcal{F}\left[\sum_{n\in\mathbb{Z}}\delta(t-nT_b)\right]&=\sum_{n\in\mathbb{Z}}e^{-j2\pi nT_bf}\\&=\sum_{n\in\mathbb{Z}}e^{j2\pi nT_bf}\end{aligned}$$

- The above equations imply that:

$$\sum_{n\in\mathbb{Z}}e^{j2\pi nT_bf}=\frac{1}{T_b}\sum_{n\in\mathbb{Z}}\delta\left(f-\frac{n}{T_b}\right)$$

This is known as the Poisson sum formula.

PSD for Unipolar NRZ Signaling (5)

- Consequently, the PSD simplifies to:

$$\begin{aligned}\mathcal{P}_s(f) &= \frac{A^2 T_b}{4}\mathrm{sinc}^2\left(T_b f\right)\left[1+\frac{1}{T_b}\sum_{n\in\mathbb{Z}}\delta\left(f-\frac{n}{T_b}\right)\right]\\ &= \frac{A^2 T_b}{4}\left[\mathrm{sinc}^2\left(T_b f\right)+\frac{1}{T_b}\mathrm{sinc}^2\left(T_b f\right)\sum_{n\in\mathbb{Z}}\delta\left(f-\frac{n}{T_b}\right)\right]\\ &= \frac{A^2 T_b}{4}\left[\mathrm{sinc}^2\left(T_b f\right)+\frac{1}{T_b}\sum_{n\in\mathbb{Z}}\mathrm{sinc}^2\left(n\right)\delta\left(f-\frac{n}{T_b}\right)\right]\end{aligned}$$

- On the other hand, given that $\mathrm{sinc}(0)=1$ and $\mathrm{sinc}(n)=0$ when n is a non-zero integer:

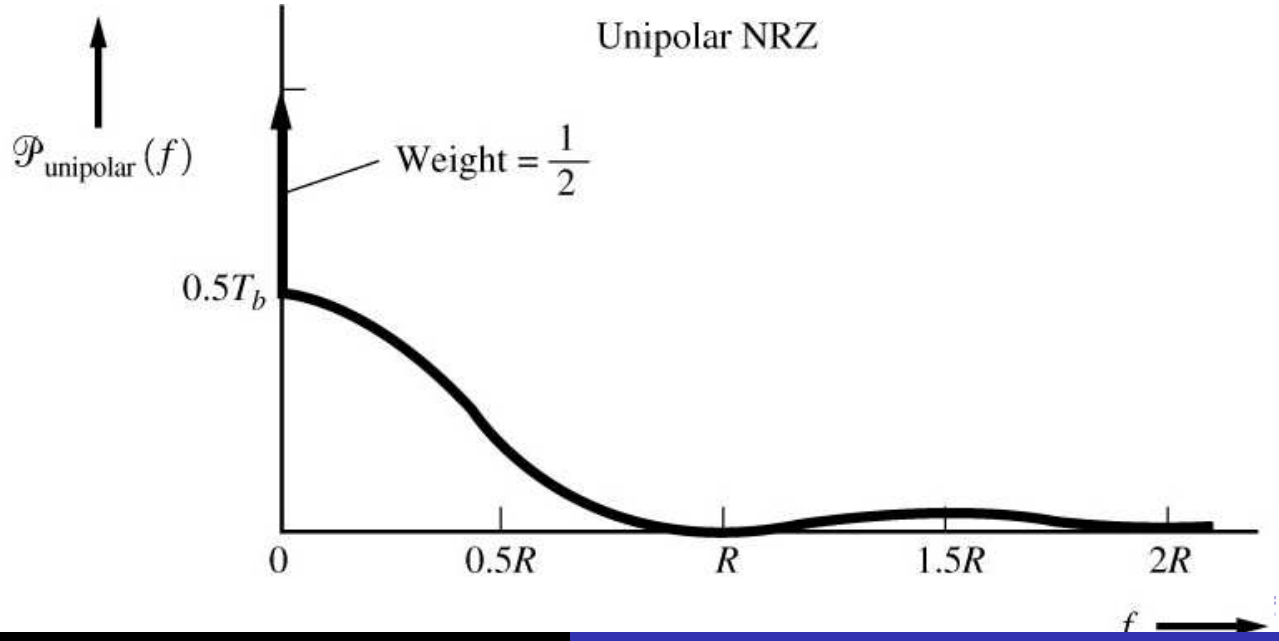
$$\begin{aligned}\mathcal{P}_s(f) &= \frac{A^2 T_b}{4}\left[\mathrm{sinc}^2\left(T_b f\right)+\frac{1}{T_b}\delta(f)\right]\\ &= \frac{A^2 T_b}{4}\mathrm{sinc}^2\left(T_b f\right)+\frac{A^2}{4}\delta(f)\end{aligned}$$

PSD for Unipolar NRZ Signaling (6)

- On the other hand, the average power of a unipolar NRZ signal is:

$$P = \frac{1}{2}\frac{1}{T_b}\int_0^{T_b} A^2 \mathrm{d}t + \frac{1}{2}\frac{1}{T_b}\int_0^{T_b} 0 \mathrm{d}t = \frac{1}{2}A^2$$

- A is selected to normalize the average power. Consequently, $P = 1$ implying that $A = \sqrt{2}$.
- The PSD for $A = \sqrt{2}$ is plotted in the next figure (where $R = \frac{1}{T_b}$).



PSD for Polar NRZ Signaling (1)

- For polar NRZ, the possible values of the a's are $+A$ and $-A$.
- For $k = 0$:

$$R(0) = \sum_{i=1}^{I=2} (a_n a_n)_i P_i = A^2 \frac{1}{2} + (-A)^2 \frac{1}{2} = A^2$$

- For $k \neq 0$:

$$\begin{aligned} R(k) &= \sum_{i=1}^{I=4} (a_n a_{n+k})_i P_i \\ &= A^2 \frac{1}{4} + (-A)A\frac{1}{4} + A(-A)\frac{1}{4} + (-A)^2 \frac{1}{4} = 0 \end{aligned}$$

- Consequently, the PSD for polar NRZ is:

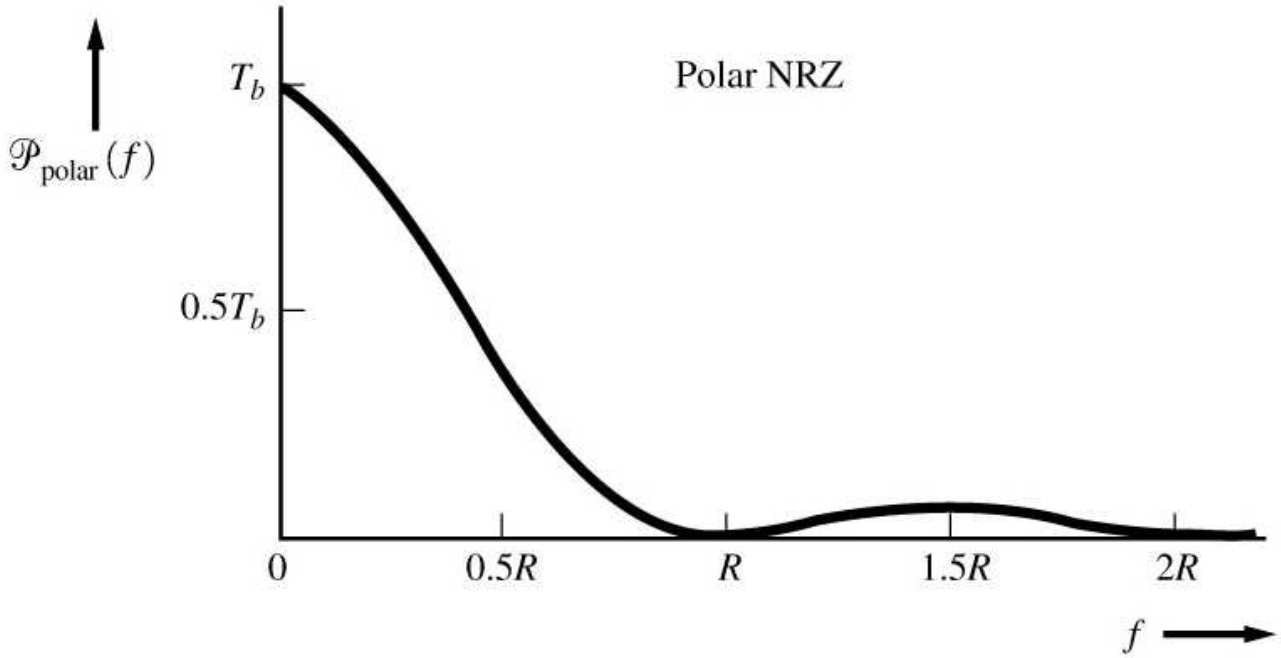
$$\begin{aligned} \mathcal{P}_s(f) &= \frac{1}{T_b} T_b^2 \mathrm{sinc}^2 \left(T_b f \right) \left[A^2 \right] \\ &= A^2 T_b \mathrm{sinc}^2 \left(T_b f \right) \end{aligned}$$

PSD for Polar NRZ Signaling (2)

- On the other hand, the average power of a polar NRZ signal is:

$$P = \frac{1}{2}\frac{1}{T_b}\int_0^{T_b} A^2 \mathrm{d}t + \frac{1}{2}\frac{1}{T_b}\int_0^{T_b} (-A)^2 \mathrm{d}t = A^2$$

- The average power is normalized when $A = 1$.
- For this value of A, the PSD is plotted in the next figure (where $R = \frac{1}{T_b}$).



PSD for Unipolar RZ Signaling (1)

- The autocorrelation for unipolar RZ is the same as the autocorrelation for unipolar NRZ. In other words:

$$R(k) = \begin{cases} \frac{1}{2}A^2, & k = 0; \\ \frac{1}{4}A^2, & k \neq 0. \end{cases}$$

- On the other hand, the rectangular pulses used with unipolar RZ have a duration of $\frac{T_b}{2}$:

$$f(t) = \mathrm{rect}\left(\frac{t}{T_b/2}\right) \quad \Rightarrow \quad F(f) = \frac{T_b}{2}\mathrm{sinc}\left(\frac{T_b}{2}f\right)$$

- $T_s = T_b$ and the PSD for unipolar RZ is:

$$\mathcal{P}_s(f) = \frac{1}{T_b}\frac{T_b^2}{4}\mathrm{sinc}^2\left(\frac{T_b}{2}f\right)\left[\frac{1}{2}A^2 + \frac{1}{4}A^2 \sum_{k=-\infty\ ,\ k\neq 0}^{+\infty} e^{j2\pi k T_b f}\right]$$

PSD for Unipolar RZ Signaling (2)

- The expression of the PSD simplifies to:

$$\mathcal{P}_s(f) = \frac{A^2 T_b}{16} \mathrm{sinc}^2\left(\frac{T_b}{2} f\right) \left[1 + \sum_{k=-\infty}^{+\infty} e^{j2\pi k T_b f}\right]$$
$$= \frac{A^2 T_b}{16} \mathrm{sinc}^2\left(\frac{T_b}{2} f\right) \left[1 + \frac{1}{T_b} \sum_{k=-\infty}^{+\infty} \delta\left(f - \frac{k}{T_b}\right)\right]$$

where the last equation follows from applying the Poisson sum formula.

- Note that:

$$\mathrm{sinc}^2\left(\frac{T_b}{2} f\right) \delta\left(f - \frac{k}{T_b}\right) = \mathrm{sinc}^2\left(\frac{k}{2}\right) \delta\left(f - \frac{k}{T_b}\right)$$

consequently, the last expression is:

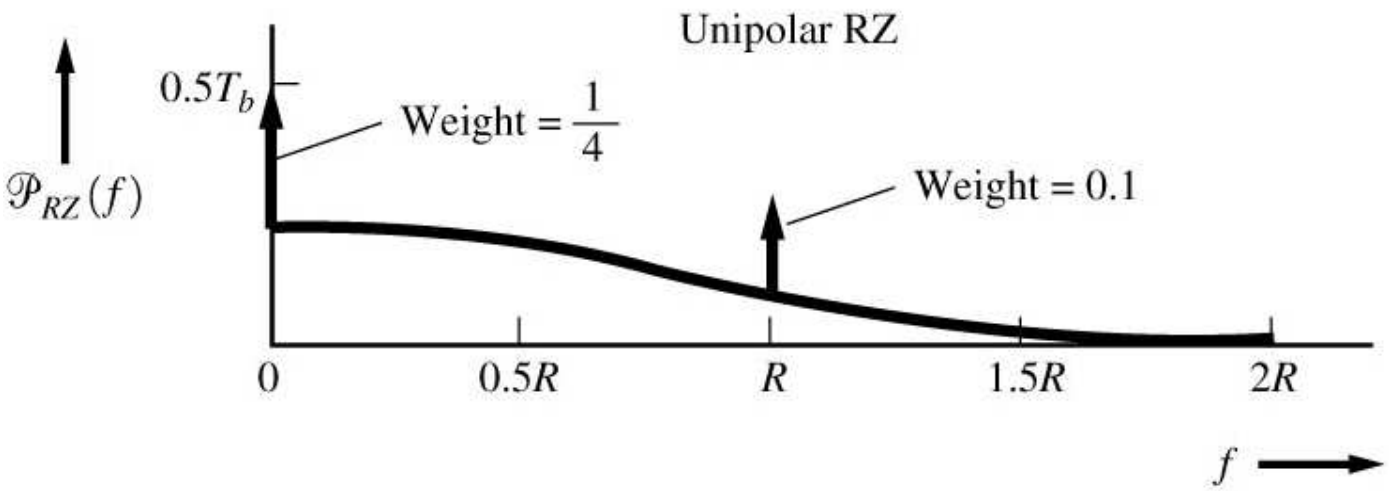
  - $\neq 0$ when $k = 0$ or when k is odd.
  - $= 0$ when $k \neq 0$ is even.

PSD for Unipolar RZ Signaling (3)

- The average power of a unipolar RZ signal is:

$$P = \frac{1}{2}\frac{1}{T_b}\int_0^{T_b/2} A^2 \mathrm{d}t + \frac{1}{2}\frac{1}{T_b}\int_0^{T_b} 0 \mathrm{d}t = \frac{1}{4}A^2$$

- The average power is normalized when $A = 2$.
- For this value of A, the PSD is plotted in the next figure (where $R = \frac{1}{T_b}$).



PSD for Unipolar RZ Signaling (4)

- The null-to-null bandwidth for unipolar RZ is twice that of unipolar or polar NRZ since the pulses are half as wide.
- The impulse term at $f = R$ can be used to recover the clock signal (this is a periodic component having a frequency R).
- The first impulse in the PSD is given by:

$$\frac{A^2}{16}\left[\mathrm{sinc}^2\left(\frac{T_b}{2}f\right)\delta\left(f-\frac{k}{T_b}\right)\right]_{k=0} = \frac{A^2}{16}\delta(f) = \frac{1}{4}\delta(f)$$

- The second impulse in the PSD is given by:

$$\begin{aligned}\frac{A^2}{16}\left[\mathrm{sinc}^2\left(\frac{T_b}{2}f\right)\delta\left(f-\frac{k}{T_b}\right)\right]_{k=1} &= \frac{A^2}{16}\mathrm{sinc}^2\left(\frac{1}{2}\right)\delta(f-R)\\ &= \frac{A^2}{16}\left[\frac{\sin\left(\frac{\pi}{2}\right)}{\frac{\pi}{2}}\right]^2\delta(f-R)\\ &= \frac{A^2}{4\pi^2}\delta(f-R) = \frac{1}{\pi^2}\delta(f-R)\end{aligned}$$

PSD for Bipolar RZ Signaling (1)

- For bipolar RZ, the symbols can take three possible values $\{0, A, -A\}$ where binary 1's are represented by alternating $+A$ and $-A$ while binary zeros are represented by a zero level.
- For $k = 0$, the autocorrelation is given by:

$$R(0) = \sum_{i=1}^{I=2} (a_n^2)_i P_i = (\pm A)^2 \frac{1}{2} + 0^2 \frac{1}{2} = \frac{1}{2} A^2$$

- For $|k| = 1$, the possible data sequences are:

$$(0,0) \ : \ (a_n, a_{n+k}) = (0,0) \text{ with probability } \frac{1}{2}.\frac{1}{2} = \frac{1}{4}$$

$$(0,1) \ : \ (a_n, a_{n+k}) = \begin{cases} (0,+A), & \text{with probability } \frac{1}{2}.\frac{1}{2}.\frac{1}{2} = \frac{1}{8}; \\ (0,-A), & \text{with probability } \frac{1}{8}. \end{cases}$$

$$(1,0) \ : \ (a_n, a_{n+k}) = \begin{cases} (+A,0), & \text{with probability } \frac{1}{8}; \\ (-A,0), & \text{with probability } \frac{1}{8}. \end{cases}$$

$$(1,1) \ : \ (a_n, a_{n+k}) = \begin{cases} (+A,-A), & \text{with probability } \frac{1}{8}; \\ (-A,+A), & \text{with probability } \frac{1}{8}. \end{cases}$$

PSD for Bipolar RZ Signaling (2)

- Consequently, for $|k| = 1$:

$$R(k) = (0)(0)\frac{1}{4} + (0)(A)\frac{1}{8} + (0)(-A)\frac{1}{8} + (A)(0)\frac{1}{8} + (-A)(0)\frac{1}{8} + (A)(-A)\frac{1}{8} + (-A)(A)\frac{1}{8} = -\frac{1}{4}A^2$$

- Now we consider the case: $|k| > 1$.
  - When either a_n or a_{n+k} are equal to zero, $a_n a_{n+k} = 0$ and the corresponding term can be removed from the expression of $R(k)$. We only need to consider the case where $(a_n, a_{n+k}) = (1, 1)$.
  - Note that $(a_n, a_{n+k}) = (1, 1)$ with probability $\frac{1}{4}$.
  - If there is an even number of binary 1's between a_n and a_{n+k}, then $a_n a_{n+k} = -A^2$.
  - If there is an odd number of binary 1's between a_n and a_{n+k}, then $a_n a_{n+k} = A^2$.
  - Since the data is independent and equiprobable, then the probability of having an odd number of 1's is the same as the probability of having an even number of 1's.

PSD for Bipolar RZ Signaling (3)

- Consequently, for $|k| > 1$:

$$R(k) = (-A^2)\frac{1}{4}.\frac{1}{2} + A^2\frac{1}{4}.\frac{1}{2} = 0$$

- Finally:

$$R(k) = \begin{cases} \frac{1}{2}A^2, & k = 0; \\ -\frac{1}{4}A^2, & |k| = 1; \\ 0, & |k| > 1. \end{cases}$$

- On the other hand, the rectangular pulses used with bipolar RZ have a duration of $\frac{T_b}{2}$:

$$f(t) = \mathrm{rect}\left(\frac{t}{T_b/2}\right) \quad \Rightarrow \quad F(f) = \frac{T_b}{2}\mathrm{sinc}\left(\frac{T_b}{2}f\right)$$

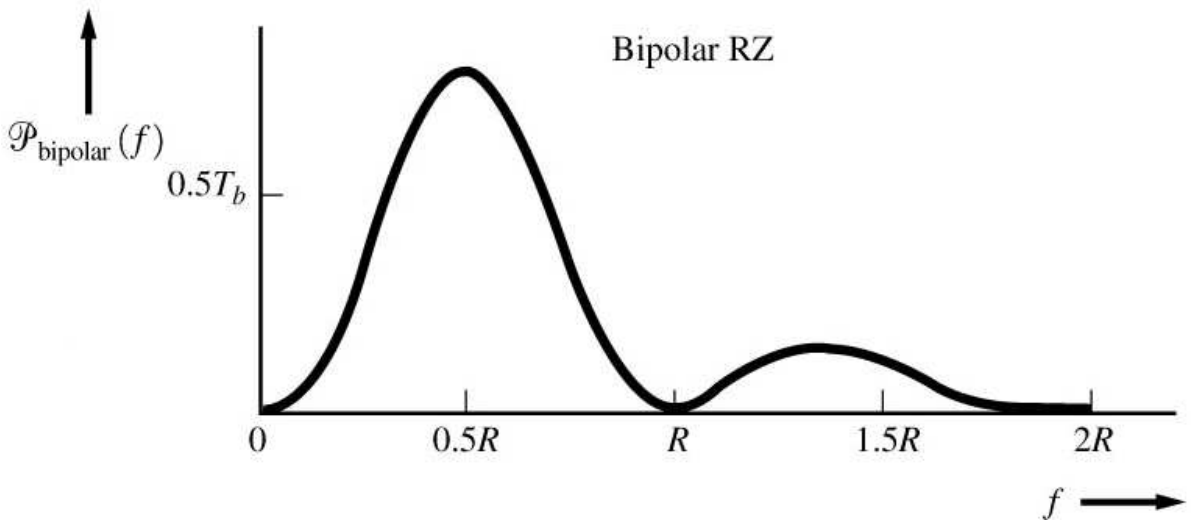
PSD for Bipolar RZ Signaling (4)

- The PSD for bipolar RZ is given by:

$$\begin{aligned}\mathcal{P}_s(f) &= \frac{1}{T_b}\frac{T_b^2}{4}\mathrm{sinc}^2\left(\frac{T_b}{2}f\right)\left[\frac{A^2}{2}-\frac{A^2}{4}\left(e^{-j2\pi T_b f}+e^{j2\pi T_b f}\right)\right]\\ &= \frac{A^2 T_b}{8}\mathrm{sinc}^2\left(\frac{T_b}{2}f\right)[1-\cos(2\pi T_b f)]\\ &= \frac{A^2 T_b}{4}\mathrm{sinc}^2\left(\frac{T_b}{2}f\right)\sin^2(\pi T_b f)\end{aligned}$$

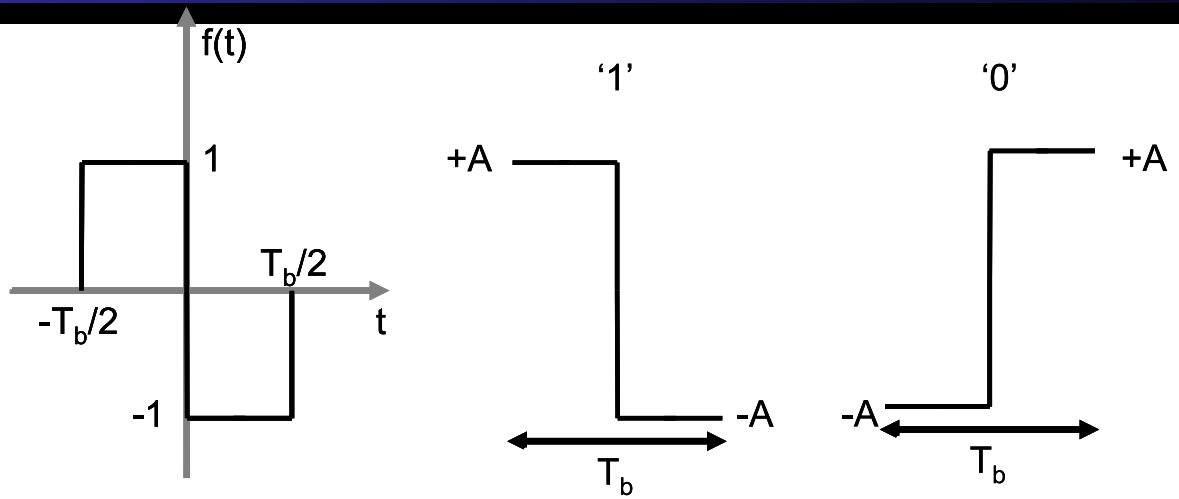
- Note that:
  - $\mathrm{sinc}\left(\frac{T_b}{2}f\right)=0$ when f is a non-zero multiple of $2R$ ($R=\frac{1}{T_b}$).
  - $\sin(\pi T_b f)=0$ when f is a multiple of R.
  - Consequently, $\mathcal{P}_s(f)=0$ at multiples of R and the null-to-null bandwidth is equal to R.
- As with unipolar RZ, the average power is $P=\frac{1}{4}A^2$ and this power is normalized for $A=2$.
- The PSD for bipolar RZ is shown in the next figure.

PSD for Bipolar RZ Signaling (5)



- The advantages of bipolar RZ are:
  - A PSD that has a small value for frequencies near $f = 0$. In other words, a DC-coupled channel is not required.
  - The null-to-null bandwidth is R.
  - A single-error detection capability: since a single error will cause a violation of the bipolar line code rule.
- The disadvantage is that the receiver must be able to distinguish between three levels ($+A$, $-A$ and 0).

PSD for Manchester NRZ Signaling (1)



- The Manchester signaling uses the pulse shape:

$$f(t) = \mathrm{rect}\left(\frac{t + T_b/4}{T_b/2}\right) - \mathrm{rect}\left(\frac{t - T_b/4}{T_b/2}\right)$$

- The Fourier transform of $f(t)$ is given by:

$$F(f) = \frac{T_b}{2}\mathrm{sinc}\left(\frac{T_b}{2}f\right)e^{j2\pi\frac{T_b}{4}f} - \frac{T_b}{2}\mathrm{sinc}\left(\frac{T_b}{2}f\right)e^{-j2\pi\frac{T_b}{4}f}$$

$$= jT_b\mathrm{sinc}\left(\frac{T_b}{2}f\right)\left[\frac{e^{j2\pi\frac{T_b}{4}f} - e^{-j2\pi\frac{T_b}{4}f}}{2j}\right] = jT_b\mathrm{sinc}\left(\frac{T_b}{2}f\right)\sin\left(\frac{\pi}{2}T_b f\right)$$

PSD for Manchester NRZ Signaling (2)

- For $k = 0$, the autocorrelation is given by:

$$R(0) = (A)(A)\frac{1}{2} + (-A)(-A)\frac{1}{2} = A^2$$

- For $k \neq 0$, the possible values of $a_n a_{n+k}$ are $A.A$, $A.(-A)$, $(-A).A$ and $(-A).(-A)$ with probability $\frac{1}{4}$ each:

$$R(k) = A^2\frac{1}{4} - A^2\frac{1}{4} - A^2\frac{1}{4} + A^2\frac{1}{4} = 0 \quad ; \quad k \neq 0$$

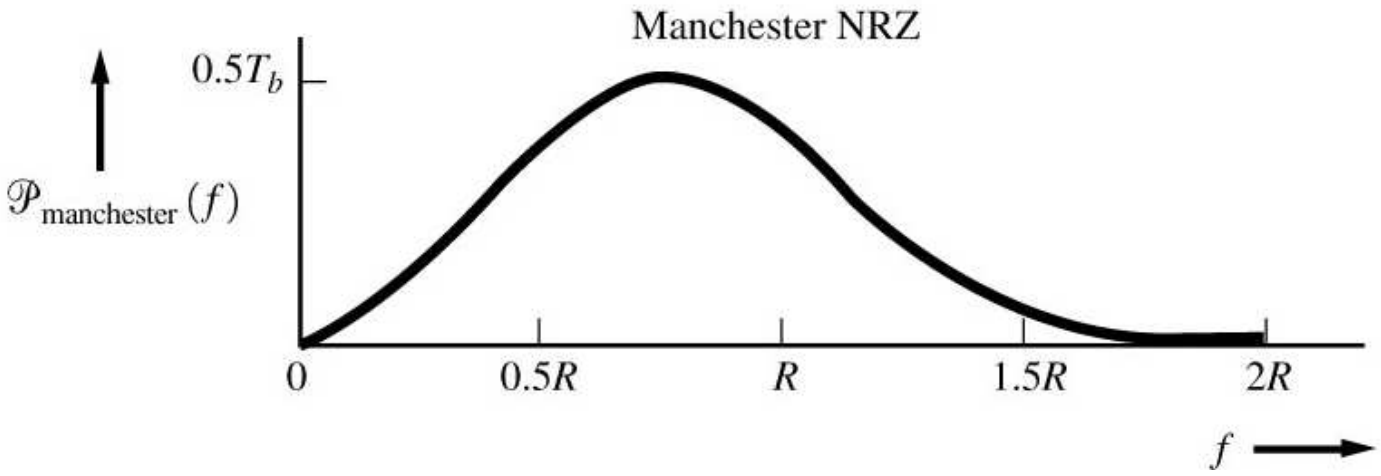
- Therefore, the PSD for Manchester signaling is:

$$\begin{aligned}\mathcal{P}_s(f) &= \frac{1}{T_b} T_b^2 \mathrm{sinc}^2\left(\frac{T_b}{2}f\right) \sin^2\left(\frac{\pi}{2}T_b f\right) \left[A^2\right] \\ &= A^2 T_b \mathrm{sinc}^2\left(\frac{T_b}{2}f\right) \sin^2\left(\frac{\pi}{2}T_b f\right)\end{aligned}$$

- The average power is $P = A^2\frac{1}{2} + (-A)^2\frac{1}{2} = A^2$ and this power is normalized for $A = 1$.

PSD for Manchester NRZ Signaling (3)

- Note that:
  - $\text{sinc}\left(\frac{T_b}{2}f\right) = 0$ when f is a nonzero multiple of $2R$ (where $R = \frac{1}{T_b}$).
  - $\sin\left(\frac{\pi}{2}T_b f\right) = 0$ when $T_b f$ is even implying that f is an even multiple of R.
  - Consequently, $\mathcal{P}_s(0) = 0$ and the null-to-null bandwidth is equal to $2R$.



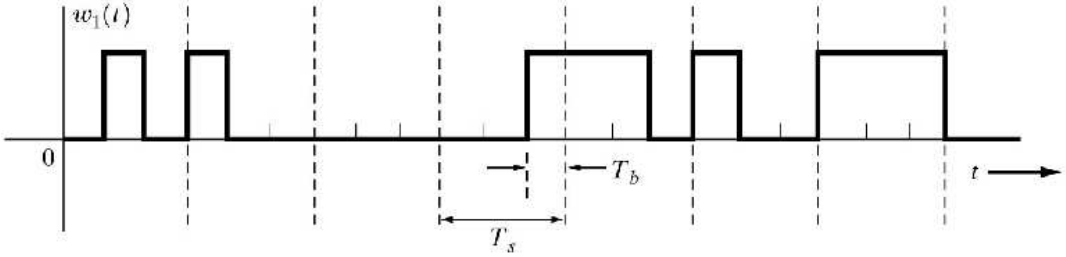
PSD for Multilevel Polar NRZ Signals (1)

- Consider the case of multilevel signaling where l bits are grouped together to form a multilevel symbol that can take $L = 2^l$ possible values.
- As an example, consider the case of $l = 3$. A possible mapping between the sequences of $l = 3$ bits and the $L = 2^l = 8$ level symbols is shown in the following table:

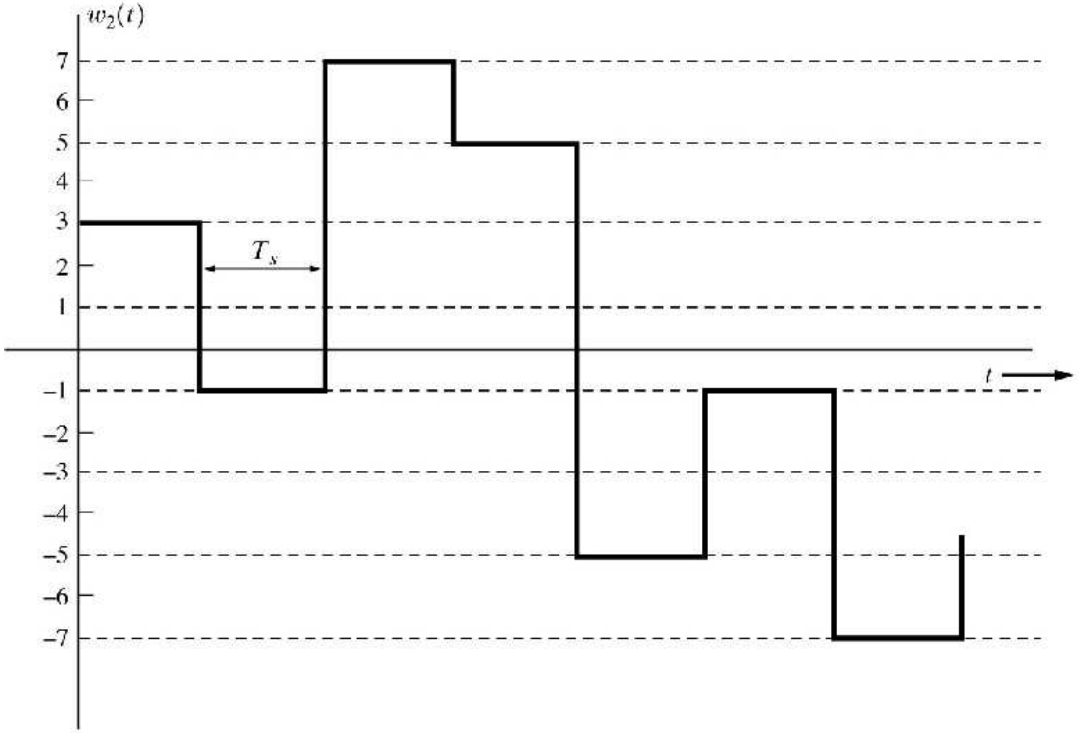
Digital word	Output level
000	+7
001	+5
010	+3
011	+1
100	-1
101	-3
110	-5
111	-7

PSD for Multilevel Polar NRZ Signals (2)

- The next figure shows an example of a 8-level signal.
- Note that $T_s = lT_b = 3T_b$.



Input Binary Waveform, $w_1(t)$



$L = 8 = 2^3$ Level Polar NRZ Waveform Out

PSD for Multilevel Polar NRZ Signals (3)

- For $k = 0$, the autocorrelation is given by:

$$R(0) = \sum_{i=1}^{8} (a_n^2)_i P_i$$
$$= \frac{1}{8}\left[7^2 + 5^2 + 3^2 + 1^2 + (-7)^2 + (-5)^2 + (-3)^2 + (-1)^2\right] = 21$$

  where $P_i = \frac{1}{8}$ for all 8 possible values $(0,0,0) \cdots (1,1,1)$.
- For $k \neq 0$, $R(k) = 0$ because of the symmetry of all levels with respect to zero.
- On the other hand, the rectangular pulse used with multilevel polar NRZ signaling is given by:

$$f(t) = \mathrm{rect}\left(\frac{t}{T_s}\right) \quad \Rightarrow \quad F(f) = T_s \mathrm{sinc}\left(T_s f\right)$$

  where $T_s = 3T_b$ (in general $T_s = lT_b$).

PSD for Multilevel Polar NRZ Signals (4)

- Consequently, the PSD for multilevel polar NRZ signaling is:

$$\begin{aligned}\mathcal{P}_s(f) &= \frac{1}{T_s} T_s^2 \mathrm{sinc}^2\left(T_s f\right) [21] \\ &= 21\, T_s \mathrm{sinc}^2\left(T_s f\right) \\ &= 63\, T_b \mathrm{sinc}^2\left(3\, T_b f\right)\end{aligned}$$

- $\mathcal{P}_s(f) = 0$ when $3T_b f$ is a nonzero integer implying that f is a nonzero multiple of $\frac{1}{3T_b} = \frac{R}{3}$ where $R = \frac{1}{T_b}$ is the bit rate.
- In other words, the null-to-null bandwidth of multilevel polar signaling is one-third the bandwidth of binary polar signaling.

PSD for Multilevel Polar NRZ Signals (5)

- In general, for $L = 2^l$ levels, the PSD of multilevel polar NRZ signals with rectangular pulse shapes is:

$$\mathcal{P}_s(f) = K \mathrm{sinc}^2 (l T_b f)$$

  where K is a constant.
- This results in a null-to-null bandwidth of:

$$B_{null} = \frac{R}{l}$$

- In summary, multilevel signaling with $L > 2$ levels is used to reduce the bandwidth of a digital signal compared with the bandwidth required for binary signaling.

Spectral Efficiency (1)

- **Spectral Efficiency** of a digital signal is given by the number of bits per second of data that can be supported by each hertz of bandwidth:

$$\eta = \frac{R}{B} \quad (\text{bits/s})/\text{Hz}$$

  where:
  - R is the data rate.
  - B is the bandwidth.
- Designers must maximize η while maintaining a certain probability of error (P_e).
- The maximum possible spectral efficiency is limited by the channel noise if the error is to be small. Following from Shannon's channel capacity formula:

$$\eta_{max} = \frac{C}{B} = \log_2\left(1 + \frac{S}{N}\right)$$

Spectral Efficiency (2)

- The spectral efficiency of the line codes is given in the following table:

Code type	First Null Bandwidth (Hz)	Spectral efficiency [(bits/s)/Hz]
Unipolar NRZ	R	1
Polar NRZ	R	1
Unipolar RZ	2R	1/2
Bipolar RZ	R	1
Manchester NRZ	2R	1/2
Multilevel Polar NRZ	R//	/